Shiham Chowdhury

Shiham | Shiham Chowdhury | shiham@umich.edu | Portfolio

Education

University of Michigan — **May 2027 (Expected)**
Bachelor of Science in Electrical Engineering — *Ann Arbor, MI*

Technical Skills

Languages: Java, Javascript, C++, Python, MATLAB
Developer Tools: Visual Studio, Jupyter Notebook, LTSpice, React, Git
Hardware: Soldering, Oscilloscope, PCB Design, Network Analyzer, Arduino, Raspberry Pi

Experience

eTitle | *Troy, MI* — May 2025 - August 2025

- Helped develop a web-based foreclosure management system designed to ease user convenience
- Tested updated merge templates using sample foreclosure cases to ensure proper functionality
- Optimized merge template scripts to remove hard-coded elements and replace outdated data

Wolverine Solar Car Team | *Ann Arbor, MI* — August 2023 - present

- Divided battery cells into multiple modules with thermistors to monitor temperature and implemented a MOSFET to switch cooling fans on when temperatures reach 40°C.
- Programmed a PID control script on an STM32 microcontroller to dynamically adjust cooling fan speeds according to battery temperature, adjusting gains to account for natural error.
- Designed a low-pass filter using MATLAB to eliminate high-frequency noise and short-term fluctuations caused by electrical interference and transient spikes.

Projects

Environment Monitoring Device | *Arduino IDE, Amazon Web Services (AWS)* — Link to Project

- Designed an IoT-based system using the ESP32 and DHT22 sensor for accurate, real-time climate data collection.
- Integrated AWS IoT Core for secure data transmission and used AWS S3 buckets for scalable and efficient storage.
- Developed a React front-end application that displays data from S3 using EC2 MATLAB scripts.

Word Occurrence Prediction Algorithm | *C++* — Link to Project

- Built a C++ algorithm that learns inputted language to predict the natural occurrence of a word.
- Utilized machine learning algorithms to analyze and categorize words into pair types for improved prediction accuracy.
- Employed recursive binary search trees to efficiently parse and manage learned data, facilitating quick computations of log-probability scores.

Euchre Game | *C++* — Link to Project

- Designed and implemented a fully functional Euchre game in C++ that supports up to 4 players.
- Integrated comprehensive game logic to enforce Euchre rules and maintain accurate game states.
- Implemented strategic decision-making algorithms based on game state to simulate CPU opponents.

Temperature-Controlled Fan | *Arduino IDE, Arduino Uno* — Link to Project

- Designed and built an Arduino-based device to control a fan's operation based on temperature readings and proximity detection.
- Utilized a DHT11 sensor to monitor temperature and humidity, with data displayed on a 16x2 LCD screen for real-time feedback.
- Implemented logic to activate DC motor-powered fan when temperature is high enough and deactivate when temperature is low or when ultrasonic sensor detects an object getting too close.